

PIZZA SALES PERFORMANCE ANALYSIS

1. Project Overview

This project analyzes pizza sales data to evaluate sales performance, product popularity, and ordering trends over time. The analysis focuses on identifying top-selling pizzas, revenue contribution by category and size, and peak sales periods to support data-driven business decisions related to product strategy and sales optimization.

2. Dataset Summary

- Rows : 48620
- Columns : 12
- Key Features :
 - Order information: order ID, order date, order time
 - Product details: pizza name, pizza category, pizza size, pizza ingredients
 - Sales metrics: quantity sold, unit price, total price
 - Transaction identifiers: pizza ID, pizza name ID
 - Missing data: no missing values detected across all columns

3. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SQL Server to answer key business questions:

- a. **Total Revenue** – To measure the total income generated from all pizza sales during the analysis period as a key indicator of overall business performance.

Results Messages	
	Total_Revenue
1	817860.05083847

- b. **Average Order Value** – To calculate the average revenue per order and evaluate how much each transaction contributes to total sales.

Results Messages	
	Average_Order_Value
1	38.3072623343546

- c. **Total Pizzas Sold** – To determine the total number of pizzas sold and understand overall sales volume.

Results Messages	
	Total_Pizzas_Sold
1	49574

- d. **Total Orders** – To count the total number of customer orders, providing insight into overall demand.

Results Messages	
	Total_Orders
1	21350

- e. **Average Pizzas Per Order** – To analyze the average number of pizzas purchased per order and identify customer purchasing patterns.

Results Messages	
	Average_Pizzas_Per_Order
1	2.32

- f. **Daily Trend for Total Orders** – To identify daily ordering patterns and determine which days experience the highest and lowest demand.

Results Messages		
	Order_Day	Total_Orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

- g. **Hourly Trend for Total Orders** – To analyze ordering behavior throughout the day and identify peak business hours.

Results Messages		
	Order_Hours	Total_Orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

- h. **Percentage of Sales by Pizza Category** – To evaluate the contribution of each pizza category to total sales and identify the most influential product categories.

Results Messages			
	pizza_category	Total_Sales	PCT
1	Classic	18619.4000015259	26.6779189176038
2	Chicken	16188.75	23.1952780348435
3	Veggie	17055.4000778198	24.4370162489706
4	Supreme	17929.7499866486	25.6897867985821

- i. **Percentage of Sales by Pizza Size** – To assess how different pizza sizes contribute to total sales and understand customer size preferences.

Results Messages			
	pizza_size	Total_Sales	PCT
1	L	95229.65	46.37
2	M	61159.00	29.78
3	S	45384.25	22.10
4	XL	3289.50	1.60
5	XXL	287.60	0.14

- j. **Total Pizzas Sold by Pizza Category** – To compare sales volume across pizza categories and identify which categories have the highest demand.

Results Messages		
	pizza_category	Total_Pizzas_Sold
1	Classic	14888
2	Chicken	11050
3	Veggie	11649
4	Supreme	11987

- k. **Top 5 Best Sellers by Total Pizzas Sold** – To identify the top five best-selling pizzas based on quantity sold and highlight the most popular products.

Results Messages		
	pizza_name	Total_Pizzas_Sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

- l. **Bottom 5 Worst Sellers by Total Pizzas Sold** – To identify the five lowest-selling pizzas and support decisions related to menu optimization or promotional strategies.

Results Messages		
	pizza_name	Total_Pizzas_Sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

4. Exploratory Data Analysis using Excel

We continue with data cleaning, data formatting, and data analysis in Excel:

- Data Loading:** Imported pizza sales data into Excel and verified data completeness.
- Data Cleaning & Formatting:** Standardized pizza size values (e.g., converting S to *Small*, M to *Medium*, L to *Large*) to ensure consistency across the dataset.
- Date Transformation:** Reformatted the order_date field and created a new order_day column to support daily trend analysis.
- Data Analysis:** Used pivot tables to calculate key business metrics, including total revenue, average order value, total orders, and total pizzas sold.
- Trend Analysis:** Built pivot tables to analyze daily and hourly ordering patterns.
- Summary Tables:** Created aggregated tables from pivot outputs to support dashboard visualizations.

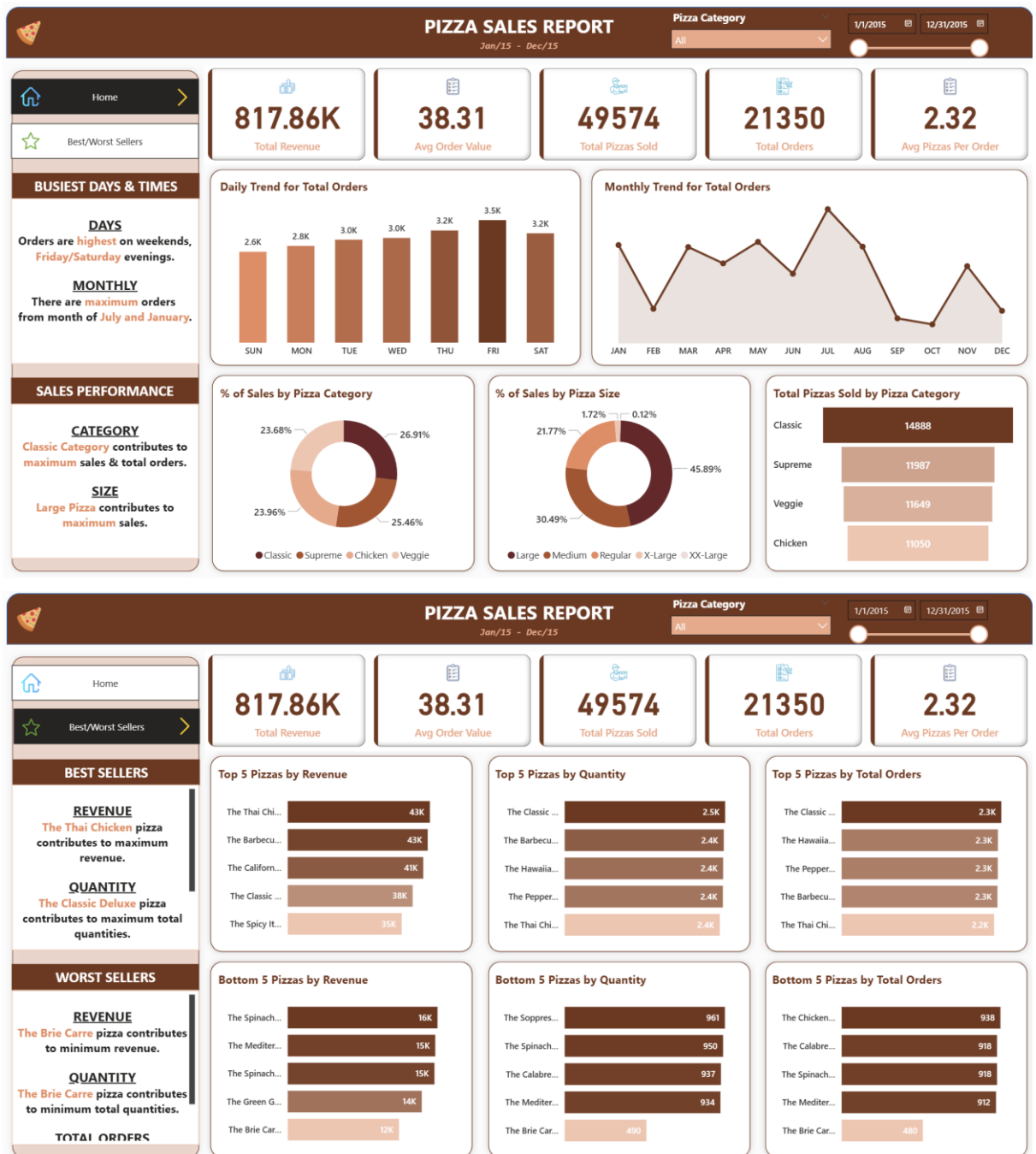
5. Dashboard in Excel

We built an interactive dashboard in **Excel** to present insights visually.



6. Dashboard in Power BI

We built an interactive dashboard in **Power BI** to present insights visually.



7. Business Recommendations

- **Optimize Weekend Promotions** – Focus marketing campaigns and special offers on Friday and Saturday evenings to maximize sales during peak demand periods.
- **Leverage High-Performing Months** – Increase marketing spend, seasonal bundles, or limited-time offers during peak months such as July and January to capitalize on higher order volumes.
- **Promote Best-Selling Categories** – Prioritize the Classic pizza category in promotions and menu placement, as it drives the highest sales and total orders.

- **Upsell Large Pizza Sizes** – Encourage customers to choose large pizzas through bundle deals or price interactive, as this size contributes the most to total sales.
- **Highlight Top Revenue Products** – Feature Thai Chicken pizza in promotional campaigns and recommendations, as it generates the highest revenue.
- **Bundle High-Volume Favorites** – Use Classic Deluxe pizza, which leads in quantity sold and total orders, as a core product in combo meals and family bundles.
- **Review Low-Performing Items** – Reassess the Brie Carre pizza, which shows the lowest performance across revenue, quantity, and orders, for potential repositioning, promotion, or removal from the menu.